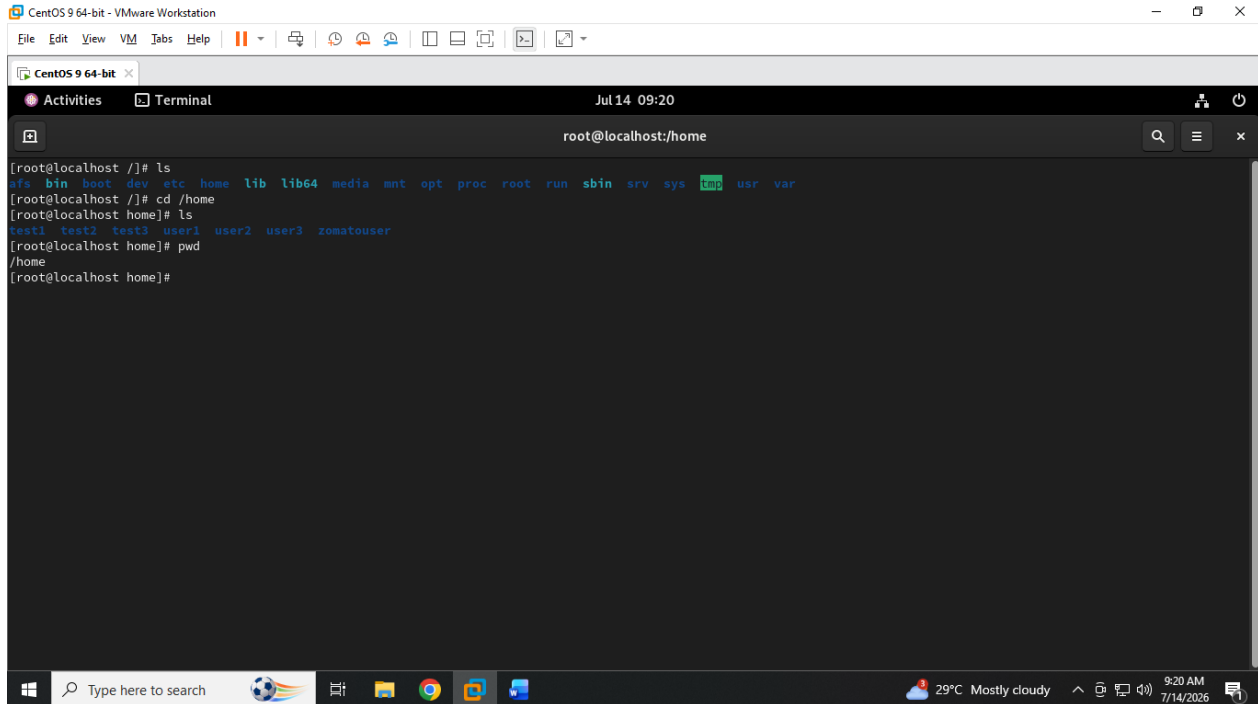


MODULE-4 : FILE SYSTEM STRUCTURE & FILE AND DIRECTORY PERMISSION SYMLINK & HARDLINK

GUIDED BY : VIPUL MISTRI

Q.1 Open a terminal and use the 'ls', 'cd', and 'pwd' commands to navigate to the /home directory, then list all user folders present. Take a screenshot of your terminal showing the commands and output.

ANS :



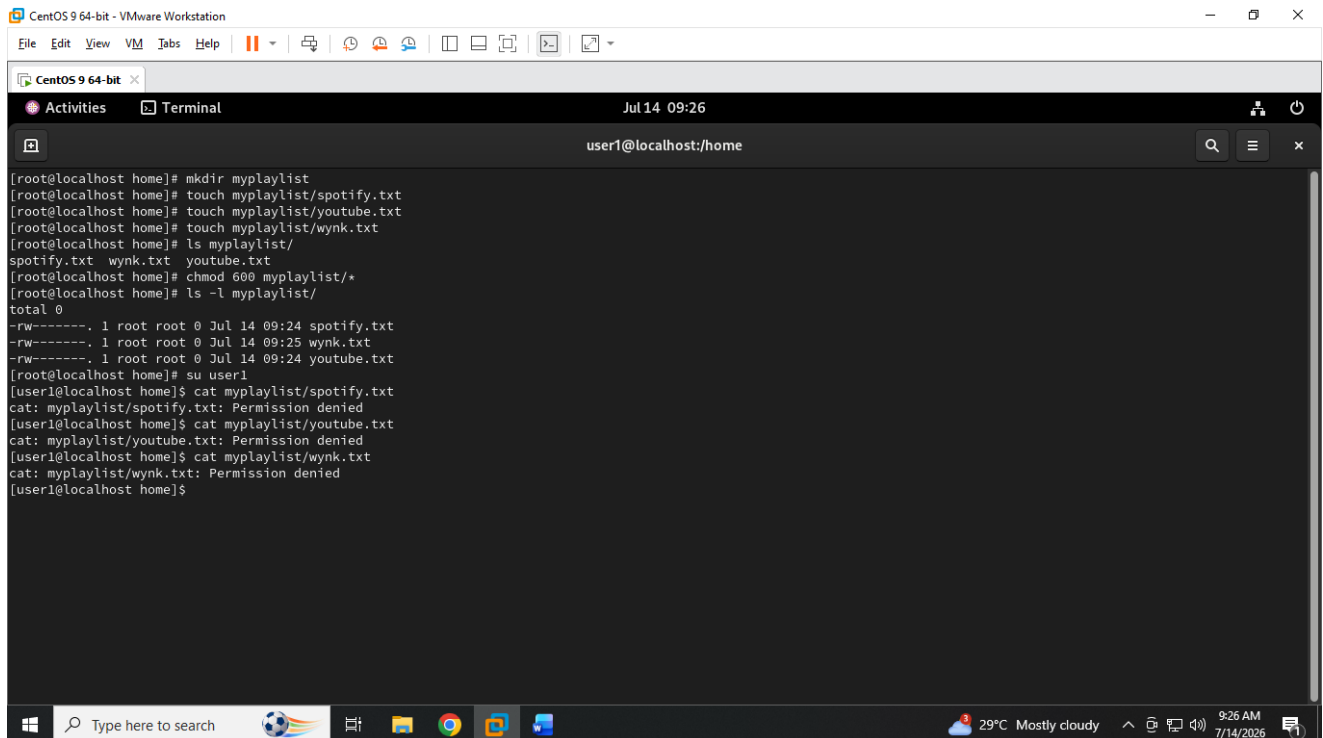
```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 14 09:20
root@localhost:/home
[root@localhost ~]# ls
afs bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys usr var
[root@localhost ~]# cd /home
[root@localhost home]# ls
test1 test2 test3 user1 user2 user3 zomatouser
[root@localhost home]# pwd
/home
[root@localhost home]#
```

MODULE-4 : FILE SYSTEM STRUCTURE & FILE AND DIRECTORY PERMISSION SYMLINK & HARDLINK

GUIDED BY : VIPUL MISTRI

Q.2 Create a new directory called 'myplaylist' inside your home folder, then use 'touch' to create three empty files named spotify.txt, youtube.txt, and wynk.txt inside it. Set the permissions so only you (the owner) can read and write these files, but others cannot access them. Use 'chmod 600' on each file after creating them.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit
Activities Terminal Jul 14 09:26 user1@localhost:/home

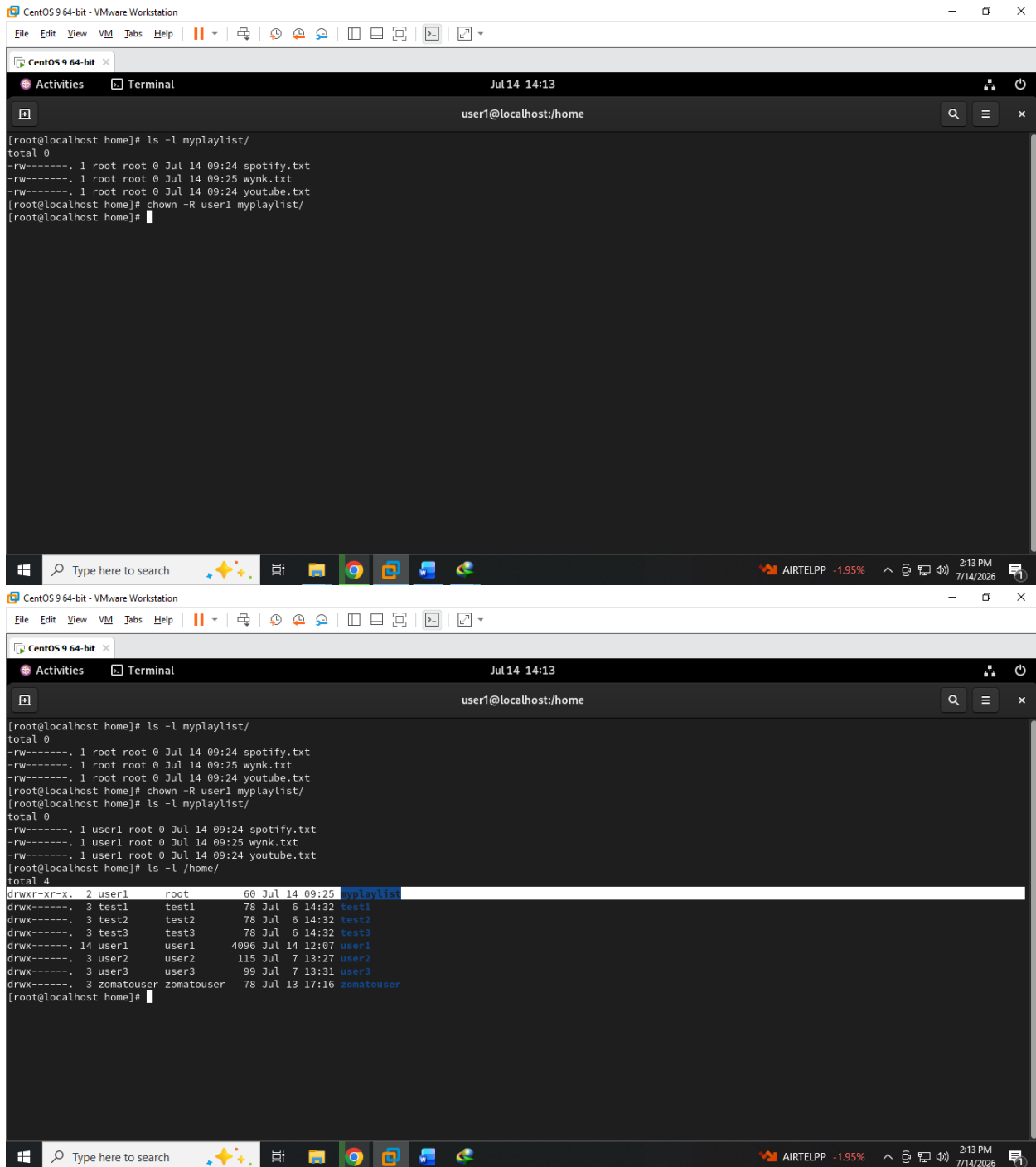
[root@localhost home]# mkdir myplaylist
[root@localhost home]# touch myplaylist/spotify.txt
[root@localhost home]# touch myplaylist/youtube.txt
[root@localhost home]# touch myplaylist/wynk.txt
[root@localhost home]# ls myplaylist/
spotify.txt wynk.txt youtube.txt
[root@localhost home]# chmod 600 myplaylist/*
[root@localhost home]# ls -l myplaylist/
total 0
-rw-----. 1 root root 0 Jul 14 09:24 spotify.txt
-rw-----. 1 root root 0 Jul 14 09:25 wynk.txt
-rw-----. 1 root root 0 Jul 14 09:24 youtube.txt
[root@localhost home]# su user1
[user1@localhost home]$ cat myplaylist/spotify.txt
cat: myplaylist/spotify.txt: Permission denied
[user1@localhost home]$ cat myplaylist/youtube.txt
cat: myplaylist/youtube.txt: Permission denied
[user1@localhost home]$ cat myplaylist/wynk.txt
cat: myplaylist/wynk.txt: Permission denied
[user1@localhost home]$
```

MODULE-4 : FILE SYSTEM STRUCTURE & FILE AND DIRECTORY PERMISSION SYMLINK & HARDLINK

GUIDED BY : VIPUL MISTRI

Q.3 Change the ownership of the 'myplaylist' directory and all its files to another user on your system using the 'chown' command. If you are the only user, create a new user first. Hint: Use 'sudo adduser newusername' to create a user if needed.

ANS :



The image shows two screenshots of a CentOS 9 64-bit terminal window. The first screenshot shows the initial state where the 'myplaylist' directory and its files are owned by root. The second screenshot shows the result after running 'chown -R user1 myplaylist/', where the ownership has been transferred to 'user1'. The terminal window also shows the output of 'ls -l /home/' which lists several users including 'user1'.

```
[root@localhost home]# ls -l myplaylist/
total 0
-rw----- 1 root root 0 Jul 14 09:24 spotify.txt
-rw----- 1 root root 0 Jul 14 09:25 wynk.txt
-rw----- 1 root root 0 Jul 14 09:24 youtube.txt
[root@localhost home]# chown -R user1 myplaylist/
[root@localhost home]#
```

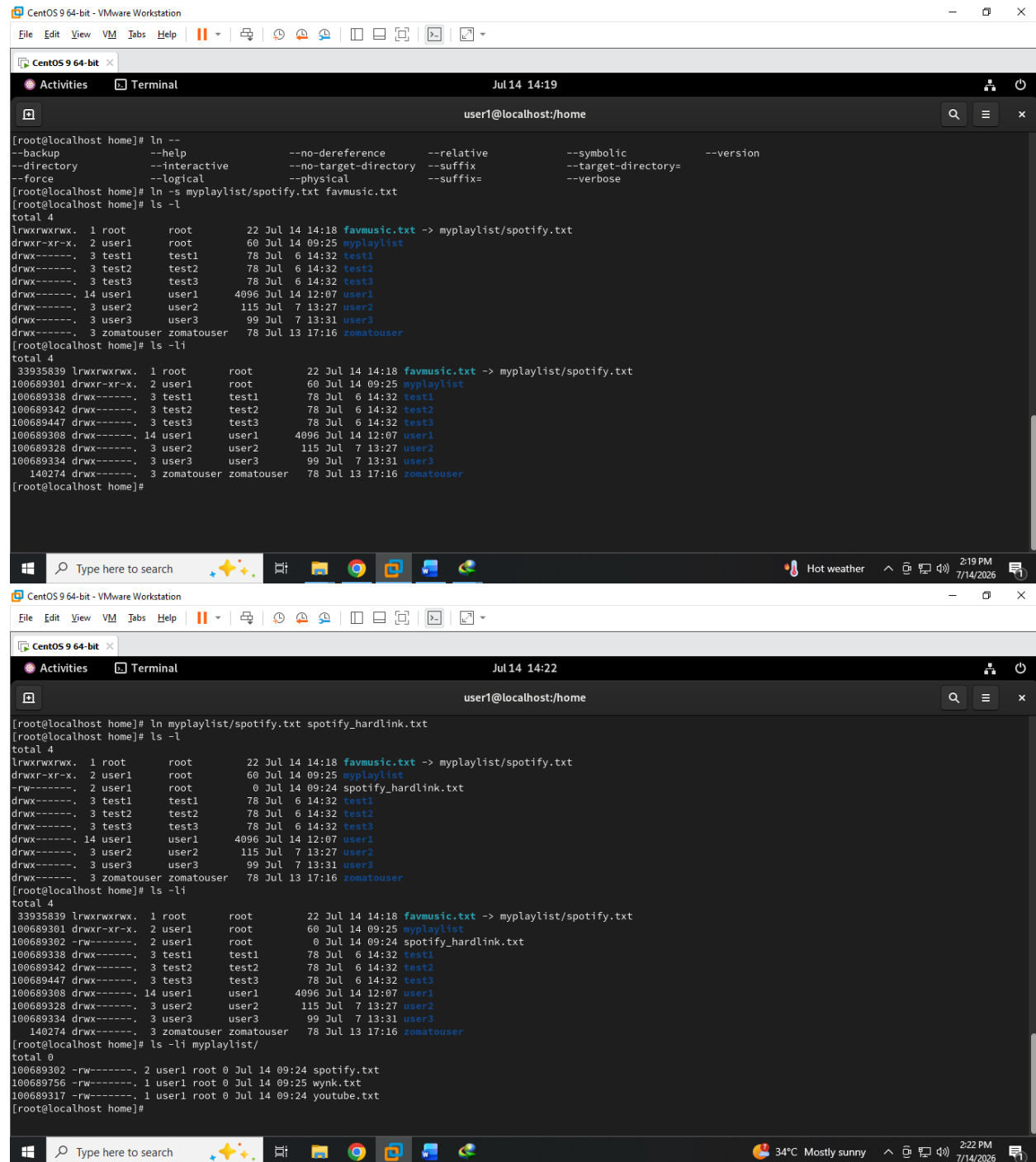
```
[root@localhost home]# ls -l myplaylist/
total 0
-rw----- 1 user1 root 0 Jul 14 09:24 spotify.txt
-rw----- 1 user1 root 0 Jul 14 09:25 wynk.txt
-rw----- 1 user1 root 0 Jul 14 09:24 youtube.txt
[root@localhost home]# ls -l /home/
total 4
drwxr-xr-x. 2 user1 root    60 Jul 14 09:25 myplaylist
drwx----- 3 test1 test1  78 Jul 6 14:32 test1
drwx----- 3 test2 test2  78 Jul 6 14:32 test2
drwx----- 3 test3 test3  78 Jul 6 14:32 test3
drwx----- 14 user1 user1 4096 Jul 14 12:07 user1
drwx----- 3 user2 user2  115 Jul 7 13:27 user2
drwx----- 3 user3 user3   99 Jul 7 13:31 user3
drwx----- 3 zomatouser zomatouser 78 Jul 13 17:16 zomatouser
[root@localhost home]#
```

MODULE-4 : FILE SYSTEM STRUCTURE & FILE AND DIRECTORY PERMISSION SYMLINK & HARDLINK

GUIDED BY : VIPUL MISTRI

Q.4 Inside your home directory, create a symbolic link named 'favmusic' that points to the spotify.txt file you created earlier. Then, create a hard link named 'spotify_hardlink' to the same file. Show the output of 'ls -l' and 'ls -li' to demonstrate the difference between the links.

ANS :



The first screenshot shows the initial state of the file system. The user is in the home directory and runs `ln --help` to see the options for creating links. Then, they create a symbolic link `favmusic` pointing to `myplaylist/spotify.txt` using `ln -s myplaylist/spotify.txt favmusic.txt`. The `ls -l` command shows the symbolic link `favmusic.txt` with permissions `lrwxrwxrwx`. The `ls -li` command shows the same file with its inode `33935839`.

```
[root@localhost home]# ln --help
--backup          --no-dereference  --relative        --symbolic        --version
--directory      --interactive    --no-target-directory --suffix          --target-directory=
--force          --logical        --physical        --suffix=         --verbose

[root@localhost home]# ln -s myplaylist/spotify.txt favmusic.txt
[root@localhost home]# ls -l
total 4
lrwxrwxrwx. 1 root    root      22 Jul 14 14:18 favmusic.txt -> myplaylist/spotify.txt
drwxr-xr-x. 2 user1   root      60 Jul 14 09:25 myplaylist
drwx----- 3 test1   test1    78 Jul 6 14:32 test1
drwx----- 3 test2   test2    78 Jul 6 14:32 test2
drwx----- 3 test3   test3    78 Jul 6 14:32 test3
drwx----- 14 user1   user1   4096 Jul 14 12:07 user1
drwx----- 3 user2   user2    115 Jul 7 13:27 user2
drwx----- 3 user3   user3     99 Jul 7 13:31 user3
drwx----- 3 zomatouser zomatouser 78 Jul 13 17:16 zomatouser
[root@localhost home]# ls -li
total 4
33935839 lrwxrwxrwx. 1 root    root      22 Jul 14 14:18 favmusic.txt -> myplaylist/spotify.txt
100689301 drwxr-xr-x. 2 user1   root      60 Jul 14 09:25 myplaylist
100689338 drwx----- 3 test1   test1    78 Jul 6 14:32 test1
100689342 drwx----- 3 test2   test2    78 Jul 6 14:32 test2
100689447 drwx----- 3 test3   test3    78 Jul 6 14:32 test3
100689308 drwx----- 14 user1   user1   4096 Jul 14 12:07 user1
100689328 drwx----- 3 user2   user2    115 Jul 7 13:27 user2
100689334 drwx----- 3 user3   user3     99 Jul 7 13:31 user3
140274 drwx----- 3 zomatouser zomatouser 78 Jul 13 17:16 zomatouser
[root@localhost home]#
```

The second screenshot shows the creation of a hard link. The user runs `ln myplaylist/spotify.txt spotify_hardlink.txt`. The `ls -l` command now shows both `favmusic.txt` (symbolic link) and `spotify_hardlink.txt` (hard link) with permissions `-rw-rw-rw-`. The `ls -li` command shows that both files share the same inode `100689302`, confirming they are hard links to the same file.

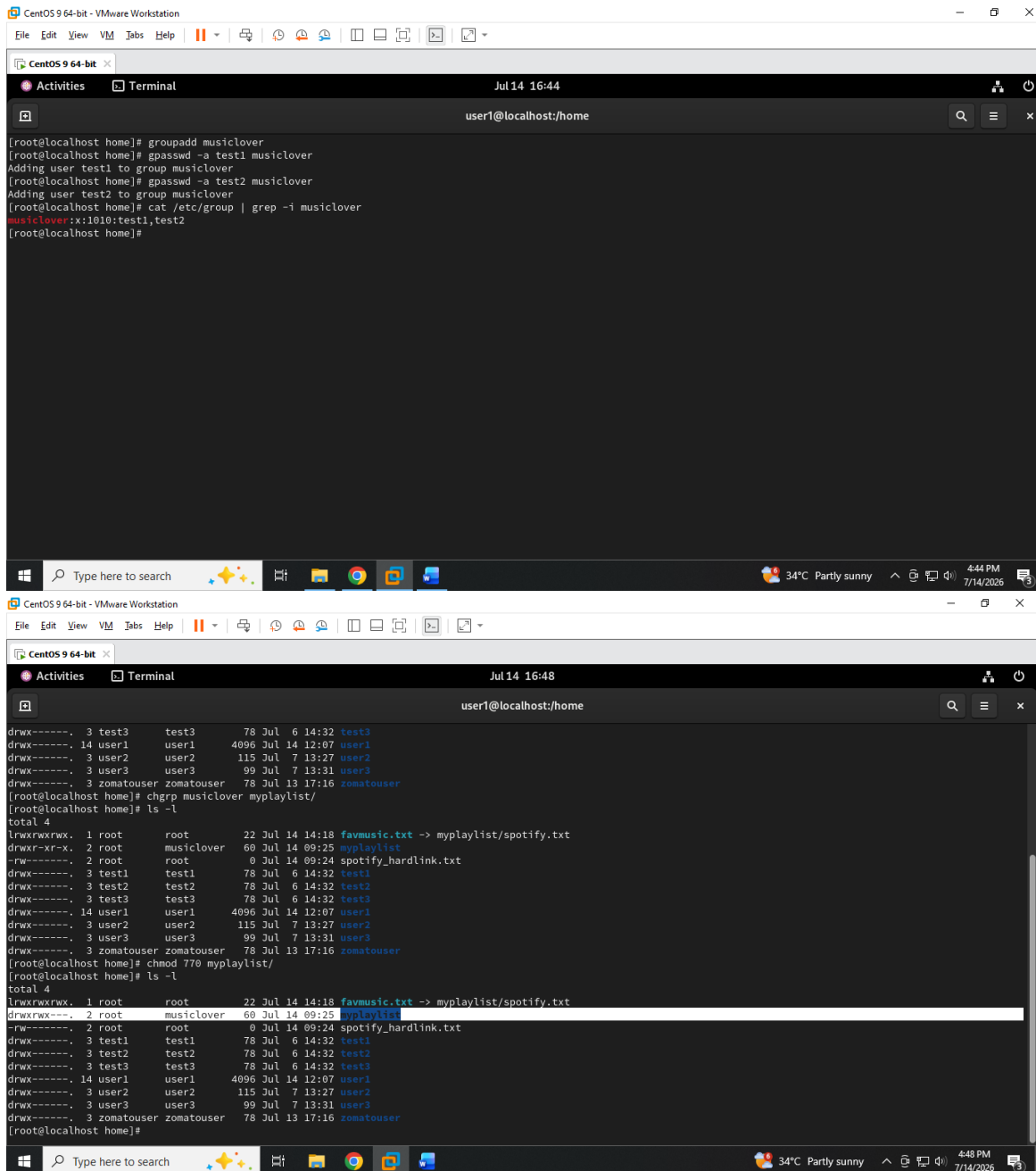
```
[root@localhost home]# ln myplaylist/spotify.txt spotify_hardlink.txt
[root@localhost home]# ls -l
total 4
lrwxrwxrwx. 1 root    root      22 Jul 14 14:18 favmusic.txt -> myplaylist/spotify.txt
-rw-rw-rw-. 2 user1   root      0 Jul 14 09:24 spotify_hardlink.txt
drwxr-xr-x. 2 user1   root      60 Jul 14 09:25 myplaylist
drwx----- 3 test1   test1    78 Jul 6 14:32 test1
drwx----- 3 test2   test2    78 Jul 6 14:32 test2
drwx----- 3 test3   test3    78 Jul 6 14:32 test3
drwx----- 14 user1   user1   4096 Jul 14 12:07 user1
drwx----- 3 user2   user2    115 Jul 7 13:27 user2
drwx----- 3 user3   user3     99 Jul 7 13:31 user3
drwx----- 3 zomatouser zomatouser 78 Jul 13 17:16 zomatouser
[root@localhost home]# ls -li
total 4
33935839 lrwxrwxrwx. 1 root    root      22 Jul 14 14:18 favmusic.txt -> myplaylist/spotify.txt
100689301 drwxr-xr-x. 2 user1   root      60 Jul 14 09:25 myplaylist
100689302 -rw-rw-rw-. 2 user1   root      0 Jul 14 09:24 spotify_hardlink.txt
100689338 drwx----- 3 test1   test1    78 Jul 6 14:32 test1
100689342 drwx----- 3 test2   test2    78 Jul 6 14:32 test2
100689447 drwx----- 3 test3   test3    78 Jul 6 14:32 test3
100689308 drwx----- 14 user1   user1   4096 Jul 14 12:07 user1
100689328 drwx----- 3 user2   user2    115 Jul 7 13:27 user2
100689334 drwx----- 3 user3   user3     99 Jul 7 13:31 user3
140274 drwx----- 3 zomatouser zomatouser 78 Jul 13 17:16 zomatouser
[root@localhost home]# ls -li myplaylist/
total 0
100689302 -rw-rw-rw-. 2 user1 root 0 Jul 14 09:24 spotify.txt
100689756 -rw-rw-rw-. 1 user1 root 0 Jul 14 09:25 wynk.txt
100689317 -rw-rw-rw-. 1 user1 root 0 Jul 14 09:24 youtube.txt
[root@localhost home]#
```

MODULE-4 : FILE SYSTEM STRUCTURE & FILE AND DIRECTORY PERMISSION SYMLINK & HARDLINK

GUIDED BY : VIPUL MISTRI

Q.5 Suppose you want to share your 'myplaylist' folder with a group of friends on the same Linux system. First, create a new group called 'musiclovers' using the 'groupadd' command, add your user to this group, and then change the group ownership of 'myplaylist' to 'musiclovers' using 'chgrp'. Set the permissions so that only group members can read and write files in this directory, but others cannot. Hint: Use 'chmod 770' on the directory after changing the group.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit
Activities Terminal Jul 14 16:44
user1@localhost/home

[root@localhost home]# groupadd musiclover
[root@localhost home]# gpasswd -a test1 musiclover
Adding user test1 to group musiclover
[root@localhost home]# gpasswd -a test2 musiclover
Adding user test2 to group musiclover
[root@localhost home]# cat /etc/group | grep -i musiclover
musiclover:x:1010:test1,test2
[root@localhost home]#

CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit
Activities Terminal Jul 14 16:48
user1@localhost/home

drwx----- 3 test3 test3 78 Jul 6 14:32 test3
drwx----- 14 user1 user1 4096 Jul 14 12:07 user1
drwx----- 3 user2 user2 115 Jul 7 13:27 user2
drwx----- 3 user3 user3 99 Jul 7 13:31 user3
drwx----- 3 zomatouser zomatouser 78 Jul 13 17:16 zomatouser
[root@localhost home]# chgrp musiclover myplaylist/
[root@localhost home]# ls -l
total 4
lrwxrwxrwx. 1 root root 22 Jul 14 14:18 favmusic.txt -> myplaylist/spotify.txt
drwxr-xr-x. 2 root musiclover 60 Jul 14 09:25 myplaylist
-rw----- 2 root root 0 Jul 14 09:24 spotify_hardlink.txt
drwx----- 3 test1 test1 78 Jul 6 14:32 test1
drwx----- 3 test2 test2 78 Jul 6 14:32 test2
drwx----- 3 test3 test3 78 Jul 6 14:32 test3
drwx----- 14 user1 user1 4096 Jul 14 12:07 user1
drwx----- 3 user2 user2 115 Jul 7 13:27 user2
drwx----- 3 user3 user3 99 Jul 7 13:31 user3
drwx----- 3 zomatouser zomatouser 78 Jul 13 17:16 zomatouser
[root@localhost home]# chmod 770 myplaylist/
[root@localhost home]# ls -l
total 4
lrwxrwxrwx. 1 root root 22 Jul 14 14:18 favmusic.txt -> myplaylist/spotify.txt
drwxrwxrwx. 2 root musiclover 60 Jul 14 09:25 myplaylist
-rw----- 2 root root 0 Jul 14 09:24 spotify_hardlink.txt
drwx----- 3 test1 test1 78 Jul 6 14:32 test1
drwx----- 3 test2 test2 78 Jul 6 14:32 test2
drwx----- 3 test3 test3 78 Jul 6 14:32 test3
drwx----- 14 user1 user1 4096 Jul 14 12:07 user1
drwx----- 3 user2 user2 115 Jul 7 13:27 user2
drwx----- 3 user3 user3 99 Jul 7 13:31 user3
drwx----- 3 zomatouser zomatouser 78 Jul 13 17:16 zomatouser
[root@localhost home]#
```